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Studierichting: Informatica
Oefeningenreeks: 1B nr. 37.12

$$2i = (x + yi)^2$$

$$2i = x^2 - y^2 + 2ixy$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 2 \end{cases}$$

$$\Rightarrow x^2 = y^2 \Rightarrow (x - y)(x + y) = 0$$

$$\Rightarrow x = y \quad \text{of} \quad x = -y$$

Geval 1: $x = y$

$$2xy = 2 \Rightarrow 2x^2 = 2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\Rightarrow (x, y) = (1, 1) \text{ of } (-1, -1)$$

Geval 2: $x = -y$

$$2xy = 2 \Rightarrow 2x(-x) = -2x^2 = 2 \Rightarrow x^2 = -1 \quad \text{geen reële oplossing}$$

$$\Rightarrow \text{Oplossingen: } x + iy = 1 + i \quad \text{en} \quad -1 - i$$

Dus: De vierkantswortels van $2i$ zijn $1 + i$ en $-1 - i$.

References